# Sefik EGRIBOZ (M.Sc., PMP, PSPO, ITIL)



## SOLUTIONS ARCHITECT

### Enterprise Integration | Data Platforms | AI-Enabled Solutions

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## PROFESSIONAL SUMMARY

**Key results:** **98%** reduction in reported safety events from look-alike and sound-alike identifiers, with AI in production | **99.8%** availability across **200+ integration points and 86 systems** | data delivery from next-day to **near-real-time** for **78 consuming systems** | **EUR 8M** cumulative savings

Solution and integration architect and Data & AI lead with 20+ years in the enterprise IT of a large, transaction-intensive 24/7 organization: designed, governed and delivered an integration backbone of **200+ integration points across 86 systems** (**~50M annual transactions**) on Kafka, IBM MQ and REST/SOAP APIs, from core operations to SAP finance, HR/payroll and DWH.

Data and AI end to end: an operational data store with canonical data model, master data management, ETL/ELT pipelines and data quality controls (~100K update records a day on average in peak season, 78 consuming systems across eleven downstream domains), and a graph neural network system in production for an **NP-hard** Quadratic Assignment Problem.

Sustained core operations through more than eightfold growth, from 10M to 85M passengers a year, as customer-side product owner for the core vendor operations suite, the B2B data exchange platform and planning platforms across three group companies: RFP/POC evaluations including a five-vendor RFP, technical due diligence and key-user training. A hands-on technical authority with **delivery accountability**; PMP-certified, rated **exceptional** in the DDI Manager Ready assessment.

## WORK EXPERIENCE

**Career Break and Professional Development** | **Istanbul** | **Sep 2024 – Present**

- Planned break after leaving the Turkish Airlines Group: family time, structured upskilling (**Harvard's CS50 AI**, 12 Python projects; **AWS Certified AI Practitioner**, **PSPO I** and **AWS Certified Cloud Practitioner** in 2026) and continued research on graph-based AI for safety-critical operations.

**Principal Solutions Architect** | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2023 – Sep 2024**

*Official title: Distinguished Expert, the organization's highest technical grade.*

- Cut reported safety events from look-alike and sound-alike identifiers by **98%**, from **50 to 1 per season** (sustained since), across up to ~1,500 daily peak-season operations, replacing 2010 heuristic assignments with an in-house graph-based similarity and assignment optimization system aligned with industry safety standards.
- Reframed identifier assignment as an **NP-hard** Quadratic Assignment Problem and built a scalable machine learning solution on **Graph Neural Networks (GNN)**: **Graph Attention Networks (GAT)** on Neo4j with PyTorch Geometric and feature engineering over phonetic, visual and semantic similarity.
- Reduced monthly incidents from **8 to 1** in 10 months as owner of roadmaps and technical direction for **six mission-critical operational systems** (1,000 users, ~100 concurrent, 5M annual transactions, EUR 1M budget), sequencing change through **integration dependency management** and architectural constraints.
- Ran a core operations platform replacement selection (**5 vendors, 3 shortlisted, 2 POCs**; a **EUR 1.5M** decision) and vendor evaluation of three planning platforms through RFP/RFI, vendor management and business cases, advising VP-level leadership on build-versus-buy, technology strategy and long-term R&D strategy.
- Held data mismatches at **zero** across two companies and two core systems during a group company's migration from TROYA, the group's in-house reservations and inventory platform, to a separate third-party platform.
- Ran data synchronization, cross-company schedule distribution and change management in a transition architecture.
- Established design authority (architecture review board) and design reviews bringing decisions under ISO 27001 compliance, software quality management and enterprise architecture governance, supporting regulatory audits and compliance.
- Led an **8-person cross-functional team** (project manager, 2 system analysts, 1 business analyst, 4 software engineers) that delivered **9 systems to production**, as architecture and design decision owner, with Scrum Master support as required; rated team members' performance.
- Defined the implementation approach, mentored engineers hands-on and cleared complex technical blockers.
- Shaped integration and technology strategy for next-generation operations platforms as cross-company subject matter expert for three group companies on the core scheduling and operations platform and on operations control systems.

**Senior Integration Architect** | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2021 – Jan 2023**

*Official title: Senior Expert, Integration Architect.*

- Drove the enterprise integration transformation: standardized **200+ integration points across 86 systems** (**~50M annual transactions**) under one integration governance and data governance model, with enterprise messaging and data streaming reference patterns adopted group-wide for interoperability, maintainability, observability and operational reliability.
- Shortened new-integration lead time from **one week to one day**, cut integration-related incidents by **99%** and retired **47 legacy point-to-point interfaces** by reusing the group-wide reference patterns.
- Achieved **99.8% availability** (planned maintenance included) and **zero unplanned downtime** on mission-critical platforms serving **500+ concurrent users**, leading cross-functional delivery management across internal and external integration platforms through production deployment under high availability, disaster recovery design and SLA/SLO management.
- Eliminated dispatches into restricted zones (**18** in the 12 months before go-live, **zero** since) by engineering a real-time early warning system across two vendor platforms: **72** automated checks and alarms on plans for up to **~1,500 daily** peak-season operations.
- Enabled **1.5M** affected customers a year to rebook online, even those already en route, by extending core-platform integration with the in-house reservations platform TROYA to real-time exception and disruption management, propagating operations control decisions to reservation, inventory and customer-facing systems as they happened.
- Modernized the integration landscape onto an event-driven application and data integration platform on Apache Kafka (topic design, schema registry), IBM MQ and REST/SOAP APIs, sustaining **~50M annual transactions**.
- Applied microservices, enterprise integration patterns, API-first design and API lifecycle management, including naming conventions, API versioning and backward compatibility.
- Engineered the reliability semantics behind the estate's **99.8%** availability: publish/subscribe and asynchronous processing, idempotency in consumers, retries and exponential backoff, timeouts and error recovery, dead-letter handling and poison-message processing, message ordering and durability, and replay of persisted streams for recovery.
- Owned the solution architecture artefacts that kept decisions traceable to a maintained target state architecture: solution blueprints, architecture roadmaps, High-Level and Low-Level Design (HLD/LLD), Architecture Decision Records (ADR) and non-functional requirements (NFR).
- Integrated a partner service behind an API gateway, handling API-key acquisition and token renewal on expiry.
- Raised release frequency from **1 to 4 a month** and **halved failed releases** by setting the estate's engineering practice: CI/CD architecture and DevOps release automation with Git-based (GitOps) configuration control, automated testing strategies, and performance and security testing gating releases.
- Ran **Scrum**-based agile delivery in **Jira** and **Confluence**: user stories with acceptance criteria, backlog management, sprint planning and Definition of Ready/Done.
- Caught **99%** of integration failures before users reported them and reduced mean time to detect from **one hour to one minute** with a real-time observability tool built across all live enterprise integrations.
- Based the tool on structured logging, correlation IDs and distributed tracing, and directed delivery of Grafana dashboards, metrics and alerting to support proactive issue detection.

**Lead Solutions Architect and Platform Owner** | **Turkish Airlines** | **Istanbul, Turkiye** | **Jun 2014 – Jan 2021**

*Official title: Project Manager, Flight Operations.*

- Designed and built a **Tier-0** (24/7 mission-critical) operational data store and evolved the operational plan data store (live since 2010) into a second Tier-0 platform.
- Took plan and operational-change delivery from **next-day to near-real-time** for **78 consuming systems**, replacing **70 separate data interfaces** by routing every change through both data stores, 24/7.
- Architected both as integration layers in front of two independent vendor platforms and as the authoritative **single source of truth** for enterprise reporting and downstream processing.
- Owned enterprise-wide operational data distribution for planning, operational and post-operation periods as the single integration point for **eleven downstream domains**: planning, workforce, load planning, reservation and customer-facing systems, cargo, maintenance, DWH, revenue and cost accounting, SAP finance, HR/payroll and regulatory reporting.
- Processed **~100K update records a day** on average in peak season, 24/7 and near-real-time, through source-to-target mappings, ETL/ELT pipelines and integration rules that consolidated, transformed and enriched every update across seasonal plans under data quality and metadata controls.
- Architected batch, near-real-time and real-time distribution patterns that absorbed **~22.5M** data changes a year (~450K annual flights, ~50 changes per flight) before fan-out to large-scale distributed data processing and analytics.
- Set consumer-specific horizons, from a **rolling three-day window** to a **90-day window**.
- Fed operational changes (delays, cancellations) to customer-notification systems, triggering **75M notifications a year**, and operational data to loyalty and disruption hotel systems, so all three ran on the same data as operations.
- Designed operational data and integration architecture on the ODS pattern, standardizing core operational and plan data enterprise-wide: logical data models, entities and relationships, canonical data structures, ESB pattern applied twice.
- Evolved both Tier-0 platforms for **ten years** (2014 – Sep 2024) as de facto product owner with final authority over feature prioritization and architectural evolution.

- Coordinated operational stakeholders and cross-team dependencies for two Tier-0 platforms, carrying releases through production under ITIL-aligned **Site Reliability Engineering (SRE)** practices and continuous monitoring.
- Drove technical direction, integration and platform modernization for operational systems with a **EUR 6.5M** total programme budget (planning, scheduling, operations control) as lead architect.
- Worked end to end: system and business analysis, software architecture and hands-on coding of critical components through release, operational handover and 24/7 mission-critical production under ITIL.
- Managed the vendor platform's master and reference data with the team: built and maintained the tables where users edit it and trained key users in master data management (MDM), keeping master data owned at its source.
- Built a reporting layer into both platforms on the same enriched data and fed it back into the source vendor platforms, so enterprise reporting and operations shared one version of the data.
- Kept **850 users** working on the core operations platform throughout the pandemic by coordinating, as its product owner, the infrastructure teams' VMware-based solution.

**Integration Solution Design Lead** | **Turkish Airlines** | **Istanbul, Turkiye** | **Jan 2007 – Jun 2014**

*Official title: Acting Team Leader.*

- Delivered **EUR 8M** in cumulative savings from 2012 to 2023 (**EUR 800K** in the first year) with a multi-currency cost reconciliation engine built on a data pipeline of entry/exit timestamps and segment distances.
- Led a **6-person team** through the lifecycle of **100+ integrations** (requirements and system analysis, design, development, testing, documentation), harmonizing project plans and tracking cross-functional milestones per project.
- Rated team members' performance and interviewed technical candidates as Acting Team Leader, with hiring decisions made by others.
- Architected the message integration and data mapping from core planning data to the in-house reservations platform (TROYA) under industry messaging standards, keeping inventory and reservations in step with cancellations, re-timings, equipment and routing changes, reinstated services and reason codes.
- Introduced (2007) and operated a standards-based B2B data exchange platform for recurring data interchange with **54 partners**.
- Headed solution design and delivery for operations, workforce and planning integrations across global technology vendors, building multithreaded real-time integration frameworks (2010) and contributing to an in-house Enterprise Service Bus (ESB) on Service-Oriented Architecture (SOA).
- Built a plan data store in 2010 to cut mainframe dependency: plans from a new vendor planning system landed in Oracle and the data warehouse, easing SAP integration; the efficiency team rated them far more accurate than the earlier vendor feed.
- Developed the first heuristic similarity and assignment system (phonetic, visual and semantic similarity), delivered in 2010: the groundwork later rebuilt as the graph-based AI solution once reclassified as NP-hard.
- Commended in writing in 2010 by the head of operations for operations projects, the identifier-safety project first among them, and personally by the **General Manager and CEO**.

**Software and Integration Engineer** | **Turkish Airlines** | **Istanbul, Turkiye** | **Apr 2000 – Jan 2007**

*Official title: Analyst Programmer.*

- Contributed to the first production implementation of a distributed vendor operations suite, later owned for two decades.
- Cut costs by **EUR 100K** a year by designing and building the timetable publication system from scratch: worldwide schedules and partner flights in, a PDF timetable for web download and print out, so the scheduling department began preparing the timetable books itself (Java, C, PL/SQL).
- Administered the suite at application level with 24/7 availability responsibility (upgrades and patches on AIX, database schema upgrades) and built its bi-directional real-time interfaces to the IBM mainframe on IBM MQ.
- Drove the technical work of the first major legacy modernization from day one as an implementation team member, moving scheduling and operations control off the IBM mainframe (ADABAS/Natural) to Oracle on Unix (AIX) and the vendor suite.
- Designed and built a 24/7 real-time world map tracing every company flight in the air with simulated positions: architecture decisions, mainframe integration over IBM MQ and UDP multicast (Java, Oracle, ILOG JViews Maps).
- Ran **statistical analysis** of historical data to determine the methodology for the project that calculated average passenger and baggage weights; the methodology was adopted for weight-and-balance calculations.

## TECHNICAL EXPERTISE (hands-on) & CORE COMPETENCIES

**Integration Architecture and Schema Governance:** Event-Driven Architecture, API Architecture, reference architectures, Apache Kafka with schema registry, schema versioning and backward-compatibility rules, controlled change across event-driven integrations, IBM MQ, Enterprise Service Bus (ESB), REST/SOAP APIs; GraphQL (design level).

**Data Architecture and Governance:** Enterprise Data Architecture, Data Modeling, Canonical Data Model, Operational Data Store (ODS), Data Integration Architecture, Data Pipeline Architecture, master data management (MDM), data lineage through source-to-target mappings, data quality and metadata controls.

**Data and AI Platform Architecture:** Graph data models (Neo4j), GNN-based assignment and decision support (PyTorch, PyTorch Geometric), data pipelines, MLOps, data warehouse and BI feeds, real-time analytics.

**Engineering Platforms & Languages:** Oracle DB, Docker and containerization, Linux (RHEL) and Unix administration (AIX, HP-UX), application administration on Windows Server; Languages: Python, Java, C, PL/SQL, shell scripting (Bash, KSH), YAML and JSON; Git and Jenkins; UML; multithreaded real-time and database-driven systems; 50+ applications and integrations to DWH, SAP and MRP, with incident analysis and performance tuning.

**Executive Leadership and Delivery:** Digital transformation, transformation governance, executive (C-level) stakeholder management, PMI/PMP delivery governance, operational readiness, customer consulting and workshop facilitation.

## EDUCATION

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| • Ph.D. (coursework) | Business Administration (Finance) | Yeditepe University | Istanbul |
| • M.Sc. | Management Engineering; thesis on data warehousing for mass customization | Istanbul Technical University | Istanbul |
| • B.Sc. | Computer Science Engineering | Istanbul University | Istanbul |

## LANGUAGES

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|---|---|---|
| • English | C1 | Full Professional |
| • German | A2 | Studying towards B1 |
| • Turkish | | Native |

## CERTIFICATIONS

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|---|---|---|
| • AWS Certified Cloud Practitioner (CLF-C02) | Amazon Web Services (AWS) | 2026 |
| • AWS Certified AI Practitioner (AIF-C01) | Amazon Web Services (AWS) | 2026 |
| • CS50's Introduction to Artificial Intelligence with Python (12 projects) | Harvard University (CS50) | 2026 |
| • Professional Scrum Product Owner I (PSPO I), Credential ID 1347372 | Scrum.org | 2026 |
| • AI and Cloud Foundations: OCI AI Foundations; Agentic AI Foundations; IBM SkillsBuild; PMI AI (CPMAI, GenAI) | Oracle, IBM, PMI | 2025 |
| • Docker Foundations Professional Certificate | Docker | 2025 |
| • Manager Ready (DDI) leadership simulation: exceptional rating | TAG Taurus Group | 2024 |
| • ITIL v3 Foundation, Candidate No. 990453483 | BCS, The Chartered Institute for IT | |
| • Project Management Professional (PMP), Credential ID 1225730 | PMI | |
| • Lufthansa Systems SchedConnect vendor training (two weeks) | Lufthansa Systems | |